

1 OCR Rockfest is a yearly music festival. Visitors buy tickets to attend and watch the music acts.

A computer system is created to sell tickets and store information about visitors and the music acts.

The number of different music acts on each of the three stages is stored in the array `actNumbers`.

Each index of the array represents one stage.

Index	0	1	2
Data	10	14	6

`actNumbers`

This algorithm adds up the total number of acts on all three stages.

```

01 count = 0

02 for x = 0 to 2

03 count = count + actNumbers[x]

04 next x

05 print(count)

```

i. Complete the trace table for the algorithm. You may not need to use all rows in the table.

Line number	count	x	Output

--	--	--	--

[4]

- ii. The array `actNumbers` is changed to include data for two more stages. The new array is shown:

Index	0	1	2	3	4
Data	10	14	6	9	11

`actNumbers`

The algorithm from part (i) needs refining to work for the new array. The algorithm is repeated here:

01 `count = 0`

02 `for x = 0 to 2`

03 `count = count + actNumbers[x]`

04 `next x`

05 `print(count)`

Identify the line number that will need to be changed for the new array and write the updated line of code.

Line number _____

Updated code _____

[2]

- iii. Refine line 05 in the algorithm to output "The total act count is " and then the content of the variable `count`

For example:

The total act count is 50

New line 05 _____

[1]

2 An array stores a collection of words. The array has the identifier data.

```
data = ["or", "and", "it", "when", "and", "or", "and", "it"]
```

A searching algorithm is used to find a word in the array data. The algorithm used to search for a word is shown.

```
word = input("Enter a word to find")

found = False

for i = 0 to 7

    if data[i] == word then

        found = True

    endif

next i

print(found)
```

Some words appear in the array more than once.

Refine the algorithm to:

- count the number of times a word appears in the array
- output this number.

For example, the algorithm should output 3 if the input is "and".

[4]

3 OCR Drones flies goods around the country using drones.

A pilot is allowed to fly a maximum of 9 hours per day. The pilot code and the hours flown for each flight in one day are stored in a 2D array of strings with the identifier `journeys`.

The data in `journeys` is shown.

	[0]	[1]
[0]	"SM720"	"4.5"
[1]	"GC730"	"3"
[2]	"JP670"	"2"
[3]	"GC730"	"3.5"
[4]	"CC200"	"9"
[5]	"RY320"	"12"

The value of `journeys[3,1]` is "3.5".

Create a function, `pilotValid()`, that:

- takes a pilot code as a parameter
- uses the 2D array `journeys` to calculate the number of hours that the pilot has flown
- returns `valid` if the pilot has flown 9 hours or fewer or `warning` if the pilot has flown for more than 9 hours
- uses casting where needed.

You must use either:

- OCR Exam Reference Language, or
- A high-level programming language that you have studied.

[6]

4 OCR Security Services need to identify the total number of seconds the sensors have been activated on a specific date.

The data from the database table `events` is imported into the program written in a highlevel programming language.

The program stores the data in a two-dimensional (2D) string array with the identifier `arrayEvents`

The data to be stored is shown in the table.

Date	SensorID	SensorType	Length
05/02/2023	WS2	Window	38
05/02/2023	MS1	Motion	2
06/02/2023	DS3	Door	1
06/02/2023	MS2	Motion	3
06/02/2023	MS1	Motion	2
07/02/2023	WS1	Window	24
07/02/2023	DS1	Door	1

In this table, the value of `events[1, 1]` contains "MS1".

- i. An array can only store data of one data type. Any non-string data must be converted to a string before storing in the array.

Identify the process that converts integer data to string data.

[1]

ii. Write a program that:

- asks the user to input a date
- totals the number of seconds sensors have been activated on the date input
- outputs the calculated total in an appropriate message including the date, for example:
Sensors were activated for 40 seconds on 05/02/2023

You must use **either**:

- OCR Exam Reference Language, or
- A high-level programming language that you have studied.

Blank lined paper for writing.

- 5 OCRBlocks is a game played on a 5×5 grid. Players take it in turns to place blocks on the board. The board is stored as a two-dimensional (2D) array with the identifier `gamegrid`

Fig. 6.1 shows that players A and B have placed three blocks each so far.

	0	1	2	3	4
0	A			B	
1					
2		B			
3	A		B		
4			A		

Fig. 6.1

The function `checkblock()` checks whether a square on the board has been filled. When `checkblock(4,2)` is called, the value "A" is returned.

```
function checkblock(r,c)
    if gamegrid[r,c] == "A" or gamegrid[r,c] == "B" then
        outcome = gamegrid[r,c]
    else
        outcome = "FREE"
    endif
    return outcome
endfunction
```

Write an algorithm to allow player A to select a position for their next block on the game board.

The algorithm must:

- ask the player for the position of their block on the board
- use the `checkblock()` function to check if this position is free
- if the position is free, add the letter "A" to the position chosen in the `gamegrid` array
- if the position is not free, repeat the above steps until a free position is chosen.

Blank lined paper for writing.

```
studentnames = ["Rob", "Anna", "Huw", "Emma", "Patrice", "Iqbal"]
```

Write an algorithm using iteration to:

- display the name of each student one at a time from `studentnames`
- take as input whether that student is present or absent
- display the total number of present students and number of absent students in a suitable message, after all student names have been displayed.

[6]

Created in ExamBuilder

Question			Answer/Indicative content	Marks	Guidance																																																
1		i	- Count initially set to 0 - count increases to 10, 24, 30 and no other changes - x goes from 0, 1, 2 (, 3) and no other changes. 30 output and no other output.	4	Example <table border="1"><thead><tr><th>Line number</th><th>count</th><th>x</th><th>Output</th></tr></thead><tbody><tr><td>1</td><td>0</td><td></td><td></td></tr><tr><td>2</td><td></td><td>0</td><td></td></tr><tr><td>3</td><td>10</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td>1</td><td></td></tr><tr><td>3</td><td>24</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td>2</td><td></td></tr><tr><td>3</td><td>30</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td>30</td></tr></tbody></table> Ignore line numbers - not required. Mark order of values. Values can be repeated in count and x columns, look at changes. FT for BP4 if incorrect answer calculated but output. Allow last value of count calculated to be output for the mark. Give BOD for “-”, “nothing”, etc as extra in output column. Do not accept <code>print (30)</code> or equivalent for output. <u>Examiner’s Comments</u> Trace tables are commonplace in Paper 2 and have examined before. Candidates showed better understanding of tracing a program through line by line, and this confidence was clear in candidate responses, with many able to correctly do this. This question differed slightly from historical examination questions in two ways. Firstly, this is the first time that a for loop has been asked about in a trace table. The challenge of the counter variable being updated was handled with ease by most candidates. Secondly, although line numbers were asked for, these were not used in the marking of candidates’ responses and instead one mark was given per set of changes to variables. This process worked extremely well and will be used in the marking of any future trace table questions.	Line number	count	x	Output	1	0			2		0		3	10			4				2		1		3	24			4				2		2		3	30			4				5			30
Line number	count	x	Output																																																		
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Question			Answer/Indicative content	Marks	Guidance
		ii	• Line 02 • For x = 0 to 4	2	Must use x as counter variable. Accept rewrites in other languages BOD 0 to 5 / range(5) etc Accept other changes that implement the change successfully e.g. using length of actNumbers <u>Examiner's Comments</u> Candidates were able to confidently identify line 02 as the line to be changed and also identified the upper bound of the for loop as the value to be changed. Where candidates provided other solutions that logically worked, these were of course accepted.

Question			Answer/Indicative content	Marks	Guidance
		iii	• <code>print("The total act count is " + count)</code>	1	Allow other concatenation operators e.g. & (VB), dot (PHP, Rust), double dot (Lua), etc. Allow comma. Must have some form of separator. Ignore spacing. Ignore case. Ignore casting. Allow other commands that are equivalent to print (e.g. echo, output, etc) Allow multiple print statements. Allow multiple lines of code that achieve the outcome. Allow string interpolation or language specific examples that will achieve this (e.g. f strings in Python, use of \$ before variables inside strings in PHP). Both of the following and examples like this are OK : • <code>print("The total act count is {count}")</code> • <code>print("The total act count is \$count")</code> <u>Examiner's Comments</u> As with many other smaller programming questions on this paper, candidates performed exceptionally well. The majority of candidates were able to use the count variable to output the correct value. Where this was not successfully done, candidates tended to include the variable identifier inside the string delimiters, thereby outputting the word count and not the value held in the variable. Examiners were instructed to credit responses that used string interpolation, such as f strings in Python or the use of \$ before strings in PHP.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
2			- Declares variable to store the count (before the for loop) - Initialises variable to zero - Increments variable by 1 each time the word is found - Outputs count value	4 (AO3 2c)	<u>Example answer :</u> <pre>search = input("Enter a word") count = 0 for i = 0 to 7 if data[i] == search then count = count + 1 next i print(count)</pre> Allow answers that completely rewrite the algorithm as long as points on left met.
			Total	4	
3			One mark per bullet point to max 6 - Function header <code>pilotValid()</code> with (at least one) parameter - Checks array element <code>[i,0]</code> for each item... ...calculates total ...casting to float / real - Checks if 9 hours or fewer Returns a value ...returns "warning" and "valid" correctly	6 (AO3 2b)	<u>Example answer :</u> <pre>function pilotValid(pilotCode) total = 0 status = "" for i = 0 to 5 if journeys[i,0] == pilotCode then temp = float(journeys[i,1]) total = total + temp endif next i if total > 9 then status = "warning" else status = "valid" endif return status endfunction</pre> Do not allow casting to integer for BP4, data shows some journeys have decimal places. Allow FT for BP5 if attempt made at calculation. Allow FT for BP7 if value is output instead of returned.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
4		i	1 mark for: • Casting / cast	1 (AO3 2a)	Accept type casting Do not accept conversion. Do not accept examples of casting. <u>Examiner's Comments</u> The use of the term "casting" to convert one data type to another is now well known and understood by candidates. This is given and referred to in the J277 specification and is essential knowledge.
		ii	1 mark each to max 6 • Input date and store in variable / use directly • Access all seven (indexes 0 to 6) events in array // loop for each event in array • Attempt at selection... • ...to compare date input against date <u>in array</u> (element 0) • ...adding length (element 3) <u>from array</u> to the total <u>if dates match</u>. • Outputting <u>calculated</u> total and date in appropriate message(s) <u>at the end</u>	6 (AO3 2b)	BP2 can be achieved either by iteration accessing each event or manually repeating code to access each event. Must be 0 to 6, not 1 to 7. Allow reference to events (table given) or arrayEvents (2D array) in answer as long as used consistently. BP2 loop allow off by one errors (Python), looping to array length or array length – 1. Allow for each item in array or any other suitable loop. BP4 and BP5 allow array reference as either column major or row major. Output can either be once at the end or on every iteration, as long as it is output at the end. Only give output mark if attempt made to calculate total <u>within the algorithm</u> . Do not penalise capitalisation or minor misspellings of variable names. <u>Example answer 1</u> <pre> total = 0 date = input("Please enter date") for count = 0 to events.length-1 if events[0, count] == date then total = total + events[3,count] endif next count print("There were " + total + " events on " + date) </pre>

Question	Answer/Indicative content	Marks	Guidance
			<u>Example answer 2</u> <pre>total = 0 date = input("Please enter date") for item in events: if item[0] == date then total = total + item[3] endif next count print("There were " + total + " events on " + date)</pre> <u>Examiner's Comments</u> The final question in Section B is expected to be a high demand question. The techniques required (iteration through a 2D array, selection, keeping a running total of times) are within the specification but it is acknowledged that the level of challenge was high. Examiners were instructed to give marks for an attempt at a solution (as with previous questions). For this question marks were given for: • any attempt at selection • any solution that accessed each element in the given array, even if this was via a manual process. Therefore, many candidates gained multiple marks for an attempt that only partially solved the problem. A significant number of candidates were able to create a solution that fully met the requirements of the question. This was often done in an elegant and efficient manner. This is extremely pleasing and shows excellent understanding and significant experience of practical programming. Exemplar 3

Question			Answer/Indicative content	Marks	Guidance
					<pre> total = 0 date = input("Enter a date") toto count = 0 for count = 0 to arrayEvents.length if arrayEvents[0, count] == date then total = total + arrayEvents [3, count] endif endfor print ("Sensors were activated for", total, "Seconds on", date) </pre> Exemplar 3 shows a high scoring response. A date has been asked for as input which has then been used to compare to each element at position 0 in the array. Where any of these match, the total variable is updated to keep a running total of the corresponding element at position 3 in the array. After each element has been checked, the total and date are output in a suitable message. This is not the only method by which a response could be given full marks but is perhaps the most common.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
5			• Input two position values separately • calls checkblock () function... • ...with input parameters • ... returned value used in selection • If free, stores "A" to correct index of gamegrid array (FT for incorrect selection) • Loops until free position chosen	6	If flowchart / structured English, do not allow simple repeat of question. <u>Example answer</u> <pre> loop = True while loop row = input("enter row") col = input("enter column") if checkblock(row,col) == "FREE" then gamegrid[row,col] = "A" loop = False endif endwhile </pre>
			Total	6	

Question	Answer/Indicative content	Marks	Guidance
6	• Use of iteration (any use) ... • ...loops for each item in array // loops 6 times • ...to print out each item in studentnames • ...input attendance • Add up/calculate students present and absent • ...Outputs present and absent (in suitable message)	6 AO3 2b(6)	BP 2 and 3 may be met together with suitable input statement. Both dependent on attempt at iteration. BP5 not dependent on correct previous parts. BP6 needs reasonable attempt at totalling present and absent figures. Ignore non-initialisation of counter variables. Flowcharts are acceptable but must show how to solve the problem, not simply repeat the question. <u>Example algorithm</u> <pre> present=0 absent=0 for i = 0 to (studentnames.length) -1 print(studentnames[i]) attendance=input("absent or present?") if attendance=="present" then present=present+1 else absent=absent+1 endif next i print ("Present students: " + present) print ("Absent students: " + absent) </pre> <u>Examiner's Comments</u> Question (c) asked candidates to write an algorithm to: 1. Call an attendance register from a given array. 2. Count and output how many students were present and absent. It was pleasing to see candidates decompose the problem and tackle each part to build a full solution. The bullet points given in the question served as a scaffold. Where one section was incorrect, other marks could be given. A number of candidates attempted to use a

Question			Answer/Indicative content	Marks	Guidance
					FOR...IN loop to iterate over the given student array. FOR...IN is not listed in the specification but is an entirely suitable way to approach the problem. Where this was logically sound, full marks were available. Other candidates used FOR...NEXT loops or WHILE loops in conjunction with the length of the array. Again, marks were given where the solution was logically sound. Marks were more limited for those candidates who did not attempt any sort of loop, but some were still able to be given where appropriate.  AfL Centres should be encouraged to connect together content within the specification, so that rather than teaching arrays in isolation, they could perhaps be combined with iteration to be able to count up or total values in an array, or apply this to how searching and sorting algorithms work. Where this is done on a regular basis, questions such as 6(c) perhaps become less daunting for candidates.  OCR support Appendix 5f of the current J276 specification covers the pseudocode guide for this examination. Candidates do not need to use this in their responses but they should be aware of this as questions will be presented using this format. For the new J277 specification, this has been replaced by OCR Exam Reference Language. This is detailed from page 25 onwards.
			Total	6	